

Unit 6

Vector spaces

Unit Goals

- Be able to explain what a vector space is, both algebraically and geometrically.
- Be able to explain how the column space of a matrix relates to the right-hand sides with solutions.
- Be able to find the special solutions to a homogenous system.
- Be able to express the general solution to a system of equations using a particular solution and the special solutions to the corresponding homogenous system.
- Be able to explain the four solution cases of a linear system using the concept of rank.

Unit 6.1

What is a vector space?

What is a vector space?

Definition in plain English:

- A vector space is a set of vectors such that all linear combinations of vectors from the set are also in the set.

Terminology:

- A mathematical set is simply a defined group of objects in which order does not matter. We generally group objects of a set by curly braces: $S = \{1,2,3\} = \{3,2,1\} = \{2,1,3\}$.

What is a vector space?

Definition in mathematical notation:

A set of vectors, V , is a vector space if and only if:

1. If $\vec{v} \in V$, then $k\vec{v} \in V$, for any scalar k .
2. If $\vec{u}, \vec{v} \in V$, then $\vec{u} + \vec{v} \in V$.

Corollaries:

3. If $\vec{u}, \vec{v} \in V$, then $k\vec{u} + l\vec{v} \in V$, for all scalars k, l ; this follows from (1) and (2).
4. The zero vector must be in the set: $\vec{0} \in V$; this follows from (1) with $k = 0$.

Notation: We read the expression, $\vec{v} \in V$, as “ $\vec{v}$ is an element of V ” or “ $\vec{v}$ is in V ”. This simply means that the vector $\vec{v}$ is a member of the set V .

What is a vector space?

Geometric interpretation in 2-D:

- If we begin with one vector: all linear combinations (i.e. scalar multiples) lie on a single line; the line passes through the origin.
- If we begin with two vectors that are not on the same line (i.e. scalar multiples): all linear combinations yield the entire 2-D space, $\mathbb{R}^2$.
- If we begin with any more than 2 vectors that are not all on the same line: all linear combinations yield the entire 2-D space, $\mathbb{R}^2$.
- If we begin with any number of vectors that are all on the same line: all linear combinations yield that line.

What is a vector space?

Geometric interpretation in 3-D:

- If we begin with one vector: all linear combinations (i.e. scalar multiples) lie on a single line; the line passes through the origin.
- If we begin with two vectors that are not on the same line: all linear combinations lie on a plane; the plane passes through the origin.
- If we begin with three vectors that are not all on the same plane: all linear combinations yield the entire 3-D space, $\mathbb{R}^3$.

What is a vector space?

Geometric interpretation in n dimensions:

- A vector space is a “flat” geometric object **that passes through the origin.**
 - A line through the origin is a vector space.
 - A plane through the origin is a vector space.
 - A hyperplane through the origin is a vector space.
- An entire n -D space, $\mathbb{R}^n$, is a special kind of vector space.
- The origin, $V_0 = \{\vec{0}\}$, is another special kind of vector space.

What is a subspace?

We say that a vector space, U , is a subspace of another, V , if (and only if):

- Every element of U is also an element of V ; $\vec{u} \in V, \forall \vec{u} \in U$.
- U is also a vector space; $\vec{u}, \vec{v} \in U$, then $k\vec{u} + l\vec{v} \in U$.

Geometrically:

- A line (through the origin) in 2-D is a subspace of $\mathbb{R}^2$.
- A line (through the origin) in 3-D is a subspace of $\mathbb{R}^3$.
- A plane (through the origin) in 3-D is a subspace of $\mathbb{R}^3$.
- The 4-D origin, $\{\vec{0} = (0,0,0,0)\}$, is a subspace of $\mathbb{R}^4$.

We often refer to $\mathbb{R}^2$ or $\mathbb{R}^3$ (or $\mathbb{R}^n$) as the “whole space” or the “parent space.”
Formally, $\mathbb{R}^n$ is a subspace of itself.

Linear spans.

- One way to define a vector space is by providing a **spanning set** of a (usually small) number of vectors.
- The **linear span** of this set is the set of all possible linear combinations of the vectors of the spanning set; it is a vector space by definition.

$$S = \{(1, 0, 1), (1, 0, -1)\}$$

$$\begin{aligned} U = \text{span}(S) &= \{k(1,0,1) + l(1,0,-1), \forall k, l \in \mathbb{R}\} \\ &= \{((k+l), 0, (k-l)) \mid \forall k, l \in \mathbb{R}\} \\ &= \{(a, 0, b) \mid \forall a, b \in \mathbb{R}\} \end{aligned}$$

$$T = \{(1, 2, 3)\}$$

$$\begin{aligned} V = \text{span}(T) &= \{k(1,2,3), \forall k \in \mathbb{R}\} \\ &= \{(k, 2k, 3k) \mid \forall k \in \mathbb{R}\} \end{aligned}$$

Abstract vector spaces.

- Vectors have two well-defined operations: addition and scalar multiplication.
 - In vector addition, we add corresponding terms. Vector addition is associative, commutative and has both an identity and an inverse:

$$\begin{aligned}\vec{u} + (\vec{v} + \vec{w}) &= (\vec{u} + \vec{v}) + \vec{w}, & \vec{u} + \vec{v} &= \vec{v} + \vec{u}, \\ \exists \vec{0}: \vec{u} + \vec{0} &= \vec{u}, \forall \vec{u}, & \forall \vec{u}, \exists -\vec{u}: \vec{u} + (-\vec{u}) &= \vec{0}\end{aligned}$$

- In scalar multiplication, we multiply all elements of the vector by the same value. Scalar multiplication is distributive with vector addition and is compatible with “regular” scalar multiplication:

$$\begin{aligned}k(\vec{u} + \vec{v}) &= k\vec{u} + k\vec{v}, & (k + l)\vec{u} &= k\vec{u} + l\vec{u}, \\ (kl)\vec{u} &= k(l\vec{u}), & 1\vec{u} &= \vec{u}, \forall \vec{u}\end{aligned}$$

- Vector spaces are **closed** with respect to these operations:

$$\vec{u} + \vec{v} \in V, \forall \vec{u}, \vec{v} \in V, \quad k\vec{v} \in V, \forall \vec{v} \in V, \forall k \in \mathbb{R}$$

- **Other mathematical objects can be worked with in the same way!**

Abstract vector spaces.

- If we can define a set of mathematical objects, V , that:

- The set has two well-defined operations, addition and scalar multiplication:
- where vector addition is associative, commutative and has both an identity and an inverse:

$$\begin{aligned} u + (v + w) &= (u + v) + w, & u + v &= v + u, \\ \exists 0 \in V: u + 0 &= u, \forall u \in V, & \forall u \in V, \exists -u \in V: u + (-u) &= 0 \end{aligned}$$

- scalar multiplication is distributive with vector addition and is compatible with “regular” scalar multiplication:

$$\begin{aligned} k(u + v) &= ku + kv, & (k + l)u &= ku + lu \\ (kl)u &= k(lu), & 1u &= u, \forall u \in V \end{aligned}$$

- and that is closed with respect to these operations:

$$u + v \in V, \forall u, v \in V, \quad ku \in V, \forall u \in V, \forall k \in \mathbb{R}$$

- then we can call V a vector space, and the elements of V are vectors.

Matrix space.

- Consider the set of all $m \times n$ matrices $\mathbb{R}^{m \times n}$.

- We can add matrices to get another matrix of the same size

$$A + B \in \mathbb{R}^{m \times n}, \forall A, B \in \mathbb{R}^{m \times n}$$

$$A + (B + C) = (A + B) + C, \quad A + B = B + A,$$

- We can define a zero matrix and an additive inverse:

$$\exists 0 \in \mathbb{R}^{m \times n}: A + 0 = A, \forall A \in \mathbb{R}^{m \times n}, \quad \forall A \in \mathbb{R}^{m \times n}, \exists -A \in \mathbb{R}^{m \times n}: A + (-A) = 0$$

- We can multiply matrices by a scalar to get another matrix of the same size

$$kA \in \mathbb{R}^{m \times n}, \forall A \in \mathbb{R}^{m \times n}, \forall k \in \mathbb{R}$$

$$k(A + B) = kA + kB, \quad (k + l)A = kA + lA$$

$$(kl)A = k(lA), \quad 1A = A, \forall A \in \mathbb{R}^{m \times n}$$

- The set of all matrices of a given size is a vector space!

Function space.

- Consider the set of all functions of a single variable, $F = \{f(x)\}$.
- We can add functions to get another function of a single variable:
$$f(x) + g(x) = h(x) \rightarrow h(x) = (f + g)(x)$$
- We can multiply functions by a scalar to get another function of a single variable:

$$kf(x) = g(x) \rightarrow g(x) = (kf)(x)$$

- Function addition: group like terms and add coefficients:
 - Obeys all the rules; the zero function is $f(x) = 0$.
- Scalar multiplication: multiply all terms by the same scalar:
 - Obeys all the rules.
- **The set of all functions of a single variable is a vector space!**

Polynomial space.

- Consider the set of all functions of a single variable that can be written as a polynomial:

$$P = \{f(x) \mid f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_n x^n\}$$

- When we add polynomials, we get another polynomial:

$$f(x) + g(x) \in P, \forall f(x), g(x) \in P$$

- When we scale polynomials, we get another polynomial:

$$kf(x) \in P, \forall f(x) \in P, \forall k \in \mathbb{R}$$

- The set of polynomials is closed with respect to addition and scalar multiplication.
- **The set of all polynomial functions of a single variable is a vector space; it is a subspace of function space!**

Unit 6.2

The column space and solvable right-hand sides.

Linear spans of rows and columns

- One way to define a vector space is by providing a **spanning set** of a (usually small) number of vectors.
- The **linear span** of this set is the set of all possible linear combinations of the vector of the spanning set.
- An $m \times n$ matrix can be thought of either as:

- m row vectors, each of length n , $A = \begin{bmatrix} (\vec{a}_1^R)^T \\ (\vec{a}_2^R)^T \\ \vdots \\ (\vec{a}_m^R)^T \end{bmatrix}$

- n column vectors, each of length m , $A = [(\vec{a}_1^C) \quad (\vec{a}_2^C) \quad \cdots \quad (\vec{a}_n^C)]$

- We can take the linear span of these sets of vectors!

The row space of a matrix.

- The row space of a matrix is the **linear span** of the vectors corresponding to each row.
- We define a spanning set from the rows:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} (\vec{a}_1^R)^T \\ (\vec{a}_2^R)^T \\ \vdots \\ (\vec{a}_m^R)^T \end{bmatrix}, (\vec{a}_i^R)^T = [a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}]$$

$$S_R = \{\vec{a}_1^R, \vec{a}_2^R, \dots, \vec{a}_m^R\}$$

- and this gives us the row space:

$$R(A) = \text{rowsp}(A) = \text{span}(S_R) = \text{span}(\{\vec{a}_1^R, \vec{a}_2^R, \dots, \vec{a}_m^R\})$$

The column space of a matrix.

- The column space of a matrix is the **linear span** of the vectors corresponding to each column.
- We define a spanning set from the columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = [(\vec{a}_1^C) \quad (\vec{a}_2^C) \quad \cdots \quad (\vec{a}_n^C)], (\vec{a}_j^C) = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

$$S_C = \{\vec{a}_1^C, \vec{a}_2^C, \cdots, \vec{a}_n^C\}$$

- and this gives us the column space:

$$\mathcal{C}(A) = \text{colsp}(A) = \text{span}(S_C) = \text{span}(\{\vec{a}_1^C, \vec{a}_2^C, \cdots, \vec{a}_n^C\})$$

The column picture of a linear system.

- A linear system can be viewed as a linear combination of columns.

$$A\vec{x} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

$$A\vec{x} = \vec{b} \rightarrow x_1(\vec{a}_1^C) + x_2(\vec{a}_2^C) + \cdots + x_n(\vec{a}_n^C) = \vec{b}$$

- A linear system $A\vec{x} = \vec{b}$ has an exact solution if and only if $\vec{b}$ can be written as a linear combination of the columns of A .

The meaning of the column space

- A linear system $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ can be written as a linear combination of the columns of A .

$$A\vec{x} = \vec{b} \rightarrow x_1(\vec{a}_1^C) + x_2(\vec{a}_2^C) + \cdots + x_n(\vec{a}_n^C) = \vec{b}$$

- The column space of A is the set of all possible linear combinations of the columns of A .

$$\mathcal{C}(A) = \text{colsp}(A) = \text{span}(\{\vec{a}_1^C, \vec{a}_2^C, \dots, \vec{a}_n^C\})$$

$$\mathcal{C}(A) = \{\vec{y} \mid \vec{y} = k_1(\vec{a}_1^C) + k_2(\vec{a}_2^C) + \cdots + k_n(\vec{a}_n^C), \forall k_i \in \mathbb{R}\}$$

- A linear system $A\vec{x} = \vec{b}$ has an exact solution if and only if $\vec{b}$ is in the column space of A .
- The column space of A is the set of all possible right-hand sides for which $A\vec{x} = \vec{b}$ has at least one exact solution.

Possible column spaces in 2-D.

- If the columns of a 2-by-n matrix are all scalar multiples of each other, the column space is a line in 2-D.

$$A = \begin{bmatrix} 2 & -4 & -2 \\ 1 & -2 & -1 \end{bmatrix} \rightarrow x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} -4 \\ -2 \end{bmatrix} + z \begin{bmatrix} -2 \\ -1 \end{bmatrix} = (x - 2y - z) \begin{bmatrix} 2 \\ 1 \end{bmatrix} = k \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- If there are (at least) two columns of a 2-by-n matrix that are not scalar multiples of each other, the column space is all of $\mathbb{R}^2$.

$$A = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix} \rightarrow x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Possible column spaces in 3-D.

- If the columns of a 3-by-n matrix are all scalar multiples of each other, the column space is a line in 3-D.
- If the columns of a 3-by-n matrix can all be expressed as linear combinations of only two of the columns (and these are not all scalar multiples of each other), the column space is a plane in 3-D.

$$\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow w \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} + z \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = (w + 2x + z) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (y + z) \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} = k \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + l \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

- If there are (at least) three columns of a 3-by-n matrix that can not be written as linear combinations of each other, the column space is all of $\mathbb{R}^3$.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

The column space of a matrix.

- The column space of a matrix is the **linear span** of the vectors corresponding to each column:

$$\mathcal{C}(A) = \text{colsp}(A) = \text{span}(\{\vec{a}_1^C, \vec{a}_2^C, \dots, \vec{a}_n^C\})$$

- The column space of A is the set of all possible right-hand sides for which $A\vec{x} = \vec{b}$ has at least one solution.
- A system $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ is in the column space of A .
- The column space of A is a subspace of $\mathbb{R}^m$, where m is the number of rows in A .
 - This includes the possibility of the column space being $\mathbb{R}^m$ itself.
 - This includes the possibility of the column space being just the zero vector, but only for one very special matrix.

Unit 6.3

The null space and the complete solution to $Ax=b$.

Homogenous systems.

- A homogenous linear system of equations is simply one with a right-hand side of all zeros:

$$A\vec{x} = \vec{0}$$

- Every matrix A can be associated with a (unique) homogenous system.
- In the row picture of a system, a homogenous system consists of planes (or hyperplanes) that all pass through the origin; their intersection is a point, line, plane, etc that also passes through the origin $\rightarrow$ a vector space!
- The origin must be a solution.
- A homogenous system always has a solution of $\vec{x} = \vec{0}$; the trivial solution.
- A homogenous system may have other solutions as well.

The null space of a matrix.

- The set of all possible solutions to the homogenous system, $A\vec{x} = \vec{0}$, is called the null space of A .

$$N(A) = \text{nullsp}(A) = \left\{ \vec{x} : A\vec{x} = \vec{0} \right\}$$

- Any linear combination of any two solutions to $A\vec{x} = \vec{0}$ must also be a solution.

$$\vec{u}, \vec{v} \in N(A) \rightarrow A\vec{u} = \vec{0}, A\vec{v} = \vec{0}$$

$$\vec{u}, \vec{v} \in N(A) \rightarrow A(k\vec{u} + l\vec{v}) = k(A\vec{u}) + l(A\vec{v}) = \vec{0}$$

$$\vec{u}, \vec{v} \in N(A) \rightarrow k\vec{u} + l\vec{v} \in N(A), \forall k, l \in \mathbb{R}$$

- **The null space is a vector space!**

The general solution to a homogenous system.

- How can we find the null space? → How do we solve a linear system?
- Forward elimination to an echelon form:

$$[A \quad \vec{0}] \rightarrow [U \quad \vec{0}]$$

- Checkpoint: If there are no free variables, there is only one solution.

$$\text{Trivial solution: } \vec{x} = \vec{0}$$

- If there are free variables, we need a general solution (in parametric form).
- Back substitution to the reduced row echelon form:

$$[A \quad \vec{0}] \rightarrow [U \quad \vec{0}] \rightarrow [R \quad \vec{0}]$$

- Solve for the pivot variables in terms of free variables, then substitute a parameter for each free variable (these can take on any value).

The general solution to a homogenous system.

• **Example:** $A\vec{x} = \vec{0}$, $A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 2 \end{bmatrix}$

$$[A \quad \vec{0}] = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 2 & 4 & 2 & 4 & 0 \\ 1 & 2 & 3 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{matrix} x_1 & +2x_2 & +2x_4 & = & 0 \\ & & x_3 & = & 0 \end{matrix}$$

$$\begin{matrix} x_1 & = & -2x_2 - 2x_4 \\ x_3 & = & 0 \end{matrix} \rightarrow (x_1, x_2, x_3, x_4) = (-2s - 2t, s, 0, t), \forall s, t \in \mathbb{R}$$

The general solution to a homogenous system.

- Alternative form:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s - 2t \\ s \\ 0 \\ t \end{bmatrix} = \begin{bmatrix} -2s \\ s \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -2t \\ 0 \\ 0 \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

- The general solution can be written as a set of all possible linear combination of vectors → a vector space

The special solutions to a homogenous system.

- The general solution to a linear system can be found directly by solving for the set of special solutions.
- A special solution has exactly one free variable set equal to 1 and all others set equal to 0.
- There are as many special solutions as there are free variables.

The special solutions to a homogenous system.

- **Example:** $A\vec{x} = \vec{0}$, $A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 2 \end{bmatrix}$

$$[A \quad \vec{0}] = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 2 & 4 & 2 & 4 & 0 \\ 1 & 2 & 3 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{matrix} x_1 & +2x_2 & +2x_4 & = & 0 \\ & & x_3 & = & 0 \end{matrix}$$

- **2 free variables $x_2, x_4 \rightarrow$ 2 special solutions:**

- $x_2 = 1, x_4 = 0 \rightarrow x_1 = -2, x_3 = 0 \rightarrow (-2, 1, 0, 0)$

- $x_2 = 0, x_4 = 1 \rightarrow x_1 = -2, x_3 = 0 \rightarrow (-2, 0, 0, 1)$

- **General solution:**

$$(x_1, x_2, x_3, x_4) = s(-2, 1, 0, 0) + t(-2, 0, 0, 1), \forall s, t \in \mathbb{R}$$

The reduced row echelon form contains the special solutions.

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 2 \end{bmatrix} \rightarrow \rightarrow \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- 2 free variables $x_2, x_4 \rightarrow$ 2 special solutions:

- $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 1 \\ ? \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \mathbf{2} & 0 & 2 \\ 0 & \mathbf{0} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} \mathbf{-2} \\ 1 \\ \mathbf{0} \\ 0 \end{bmatrix}$
- $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & \mathbf{2} \\ 0 & 0 & 1 & \mathbf{0} \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} \mathbf{-2} \\ 0 \\ \mathbf{0} \\ 1 \end{bmatrix}$

- Set one free variable to 1, others to 0.
- Select corresponding column.
- For each non-zero row:
 - The negative of the element in the free variable column is the element in the special solution corresponding to the pivot of that row.

The reduced row echelon form contains the special solutions.

- An example: $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} \end{bmatrix} \rightarrow \rightarrow \begin{bmatrix} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = R$

- 2 free variables $x_3, x_5 \rightarrow$ 2 special solutions:

$$\vec{s}_1 = \begin{bmatrix} ? \\ ? \\ 1 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} -a \\ -b \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{s}_2 = \begin{bmatrix} ? \\ ? \\ 0 \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} -c \\ -d \\ 0 \\ -e \\ 1 \end{bmatrix}$$

Possible null spaces in 2-D.

- If the rows of an m-by-2 matrix are all scalar multiples of each other, there is one free variable and the null space is a line in 2-D.

$$A = \begin{bmatrix} 2 & -4 \\ 1 & -2 \\ -3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -4 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, N(A) = \left\{ \vec{x} : \vec{x} = s \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \forall s \in \mathbb{R} \right\}$$

- If there are (at least) two rows of an m-by-2 matrix that are not scalar multiples of each other, there are no free variables and the null space is just the zero vector.

$$A = \begin{bmatrix} 2 & -4 \\ 1 & -3 \\ -2 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -4 \\ 0 & -1 \\ 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, N(A) = \{\vec{0}\}$$

Possible null spaces in 3-D.

- If the rows of an m-by-3 matrix are all scalar multiples of each other, there are two free variables and the null space is a plane in 3-D.

$$A = \begin{bmatrix} 2 & -4 & 6 \\ 1 & -2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -4 & 6 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 3 \\ 0 & 0 & 0 \end{bmatrix}, N(A) = \left\{ \vec{x}: \vec{x} = s \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}, \forall s, t \in \mathbb{R} \right\}$$

- If the rows of an m-by-3 matrix can all be expressed as linear combinations of only two of the rows (and these are not all scalar multiples of each other), there is one free variable, and the null space space is a plane in 3-D.

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, N(A) = \left\{ \vec{x}: \vec{x} = s \begin{bmatrix} -2 \\ -1 \\ 0 \end{bmatrix}, \forall s \in \mathbb{R} \right\}$$

- If there are at least three rows of an m-by-3 matrix that can not be expressed as linear combinations of the others, there are no free variables, and the null space is just the zero vector.

The null space of a matrix.

- The null space of a matrix is set of solutions to the homogenous system with that coefficient matrix:

$$N(A) = nullsp(A) = \{\vec{x}: A\vec{x} = \vec{0}\}$$

- The dimension of the null space of A is indicated by the number of free variables in an echelon form of A .
- The null space of A can be described by a set of special solutions.
- The null space of A is a subspace of $\mathbb{R}^n$, where n is the number of columns in A .
 - This includes the possibility of the null space being just the zero vector.
 - This includes the possibility of the null space being $\mathbb{R}^n$, but only for one very special matrix.

Nonhomogenous systems.

- For every nonhomogenous system, we can write an associated homogenous system:

$$A\vec{x} = \vec{b}, \quad A\vec{x} = \vec{0}$$

- If $\vec{x}_p$ is any solution to $A\vec{x} = \vec{b}$, and $\vec{x}_n$ is any solution to the homogenous system:

$$A(\vec{x}_p + \vec{x}_n) = A\vec{x}_p + A\vec{x}_n = \vec{b} + \vec{0} = \vec{b}$$

- We can add any solution of the homogenous system to a solution of the nonhomogenous system, and we still have a valid answer!

Nonhomogenous systems.

- If $\vec{x}_{p1}$ and $\vec{x}_{p2}$ are **both** solutions to $A\vec{x} = \vec{b}$:

$$A(\vec{x}_{p1} - \vec{x}_{p2}) = A\vec{x}_{p1} - A\vec{x}_{p2} = \vec{b} - \vec{b} = \vec{0}$$

- The difference of any two solutions to the inhomogenous system must be a solution to the homogenous system!
- Every solution to $A\vec{x} = \vec{b}$ is related to every other solution, by the addition of a vector from the null space of A .

$$\forall \vec{x}_{p1}, \vec{x}_{p2}: A\vec{x}_{p1} = A\vec{x}_{p2} = \vec{b}, \exists \vec{x}_n \in N(A): \vec{x}_{p2} = \vec{x}_{p1} + \vec{x}_n$$

- The general solution to $A\vec{x} = \vec{b}$ can be expressed as the sum of one particular solution and the general solution to $A\vec{x} = \vec{0}$.

A geometric interpretation.

- The set of solutions to $A\vec{x} = \vec{0}$ (the null space of A) a flat geometric object passing through the origin.
- The set of solutions to $A\vec{x} = \vec{b}$ is a flat geometric object that does not pass through the origin.
- **These two solution sets are parallel objects!**
- The set of solutions to $A\vec{x} = \vec{b}$ is the null space of A , translated away from the origin (it is an affine space, rather than a vector space).
- The general solution to $A\vec{x} = \vec{b}$ is thus $N(A) + \vec{x}_p$: the null space plus a translation vector.

The general solution to a nonhomogenous system.

- The general solution to $A\vec{x} = \vec{b}$ can be expressed as the sum of one particular solution and the general solution to $A\vec{x} = \vec{0}$.
- How to find a particular solution?
- **Gaussian Elimination on the augmented matrix!**
- Just as we can find special solutions to the homogenous system by setting all free variables but one equal to 0, we can find a special particular solution by setting every free variable equal to zero.
- **Everything we need is contained in the reduced row echelon form of the augmented matrix!**

The general solution to a nonhomogenous system.

- An example: $A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \end{bmatrix}$

$$[A \quad \vec{b}] = \begin{bmatrix} 1 & 1 & 2 & 2 & 1 \\ 2 & 2 & 1 & 1 & 5 \\ 3 & 3 & 3 & 3 & 6 \\ 1 & 1 & -1 & -1 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & 2 & 1 \\ 0 & 0 & -3 & -3 & 3 \\ 0 & 0 & -3 & -3 & 3 \\ 0 & 0 & -3 & -3 & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 2 & 2 & 1 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = [R \quad \vec{d}]$$

The general solution to a nonhomogenous system.

$$[R \quad \vec{d}] = \begin{bmatrix} 1 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- The two free variable columns contain the negative of the pivot values for the two special solutions.
- The last column contains the pivot values for the particular solution.

$$\vec{x}_{n1} = \begin{bmatrix} ? \\ 1 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} ? \\ 0 \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \vec{x}_p = \begin{bmatrix} ? \\ 0 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

The general solution to a nonhomogenous system.

- An example:

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 0 \end{bmatrix} \rightarrow \vec{x}_{n1} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \vec{x}_{n2} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \vec{x}_p = \begin{bmatrix} 3 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

- **General solution:** $S = \{ \vec{x} : \vec{x} = \vec{x}_p + s\vec{x}_{n1} + t\vec{x}_{n2} \forall s, t \in \mathbb{R} \}$

$$S = \left\{ \vec{x} : \vec{x} = \begin{bmatrix} 3 \\ 0 \\ -1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} \forall s, t \in \mathbb{R} \right\},$$

The rank of a matrix.

- The character of the column space and null space are both determined by the number of pivots (in an echelon form):
- If there are fewer pivots than rows, then some right-hand sides have not solution: the column space is not the entire parent space.
 - The number of pivots dictates the geometry of the column space.
- If there are fewer pivots than columns, then there are free variables: the null space is not the zero vector alone.
 - The number of free variables dictates the geometry of the null space.
- **Definition: The rank of a matrix is the number of pivots in an echelon form.** $r = \text{rank}(A)$

Four classes of linear system.

Consider a coefficient matrix with m rows, n columns and a rank of r .

- **Full rank (nonsingular):** If $r=m=n$: The matrix is square and non-singular (invertible); there is a unique solution for all right-hand sides.
- **Full row rank:** If $r=m$, and $n > m$: The matrix is short and wide, but every row has a pivot; there are infinite solutions for all right-hand sides.
- **Full column rank:** If $r=n$, and $m > n$: The matrix is tall and thin, but every column has a pivot; if there is a solution it will be unique, but not all right-hand sides will have a solution.
- If $r < m$ and $r < n$, regardless of the values of m and n : There are both rows and columns that do not contain a pivot; some right-hand sides will not have a solution, but those that do will have an infinite number of solutions.

Unit 6.3.1

**Worked examples with null spaces,
general solutions and rank.**

Find the special solutions.

- Find the special solutions for the following homogenous systems and use the result to describe the null space of the coefficient matrix.

$$A\vec{x} = \vec{0}, \quad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 4 \end{bmatrix}$$

$$B\vec{x} = \vec{0}, \quad B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 0 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$C\vec{x} = \vec{0}, \quad C = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 0 \end{bmatrix}$$

Find the special solutions.

- Find the special solutions for the following homogenous system and use the result to describe the null space of the coefficient matrix.

$$A\vec{x} = \vec{0}, \quad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 4 \end{bmatrix}$$

Find the special solutions.

$$A\vec{x} = \vec{0}, \quad A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 4 \end{bmatrix}$$

- First, find the reduced row echelon form:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

- There are one free variables, x_2 , and thus one special solution:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} ? \\ 1 \\ ? \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \vec{x}_n$$

- The null space is a line:

$$N(A) = \{\vec{x} : \vec{x} = s\vec{x}_n, \forall s \in \mathbb{R}\}$$

Find the special solutions.

- Find the special solutions for the following homogenous system and use the result to describe the null space of the coefficient matrix.

$$B\vec{x} = \vec{0}, \quad B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 0 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

Find the special solutions.

$$B\vec{x} = \vec{0}, \quad B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 0 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

- First, find the reduced row echelon form:

$$\begin{aligned} B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 0 & 2 \\ 2 & 2 & 1 \end{bmatrix} &\rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & -2 & 1 \\ 0 & -2 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R \end{aligned}$$

- There are no free variables, the null space is the origin, $N(B) = \{\vec{0}\}$, and there are no special solutions.

Find the special solutions.

- Find the special solutions for the following homogenous system and use the result to describe the null space of the coefficient matrix.

$$C\vec{x} = \vec{0}, \quad C = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 0 \end{bmatrix}$$

Find the special solutions.

$$C\vec{x} = \vec{0}, \quad C = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 0 \end{bmatrix}$$

- First, find the reduced row echelon form:

$$C = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 4 \\ 1 & 2 & 3 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

- There are two free variables, x_2 and x_4 , and thus two special solutions:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 1 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \vec{x}_{n2}$$

- The null space is a plane:

$$N(A) = \{\vec{x} : \vec{x} = s\vec{x}_{n1} + t\vec{x}_{n2}, \forall s, t \in \mathbb{R}\}$$

Find the general solution.

Find the general solution in the form:

$$\vec{x} = \vec{x}_p + s_1 \vec{x}_{n1} + s_2 \vec{x}_{n2} + \cdots + s_{(n-r)} \vec{x}_{N,(n-r)}$$

$$(a) \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

Find the general solution.

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

- First, find the reduced row echelon form **of the augmented matrix**:

$$\begin{aligned} [A \quad \vec{b}] &= \begin{bmatrix} 1 & 1 & 2 & 2 & 0 \\ 2 & 2 & 1 & 1 & 2 \\ 3 & 3 & 3 & 3 & 2 \\ 1 & 1 & -1 & -1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & 2 & 0 \\ 0 & 0 & -3 & -3 & 2 \\ 0 & 0 & -3 & -3 & 2 \\ 0 & 0 & -3 & -3 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & 2 & 0 \\ 0 & 0 & -3 & -3 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 & 1 & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 & -2/3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 0 & 4/3 \\ 0 & 0 & 1 & 1 & -2/3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow [R \quad \vec{d}] \end{aligned}$$

Find the general solution.

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

- Next, find the special solutions of $A\vec{x} = \vec{0}$ and a particular solution of $A\vec{x} = \vec{b}$:

$$[R \quad \vec{d}] = \begin{bmatrix} 1 & 1 & 0 & 0 & 4/3 \\ 0 & 0 & 1 & 1 & -2/3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- There are two free variables, x_2 and x_4 , and thus two special solutions:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 1 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} = \vec{x}_{n2} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} 4/3 \\ 0 \\ -2/3 \\ 0 \end{bmatrix} = \vec{x}_p$$

Find the general solution.

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 3 & 3 & 3 \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

- Finally, put it all together:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 1 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} = \vec{x}_{n2} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} ? \\ 0 \\ ? \\ 0 \end{bmatrix} = \begin{bmatrix} 4/3 \\ 0 \\ -2/3 \\ 0 \end{bmatrix} = \vec{x}_p$$

- The general solution is:

$$\vec{x} = \vec{x}_p + s\vec{x}_{n1} + t\vec{x}_{n2}, \forall s, t \in \mathbb{R}$$

$$\vec{x} = \begin{bmatrix} 4/3 \\ 0 \\ -2/3 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4/3 - s \\ s \\ -2/3 - t \\ t \end{bmatrix}$$

Find the general solution.

Find the general solution in the form:

$$\vec{x} = \vec{x}_p + s_1 \vec{x}_{n1} + s_2 \vec{x}_{n2} + \cdots + s_{(n-r)} \vec{x}_{N,(n-r)}$$

$$(b) \begin{bmatrix} 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Find the general solution.

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

- First, find the reduced row echelon form **of the augmented matrix**:

$$[A \quad \vec{b}] = \begin{bmatrix} 0 & 1 & 2 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & -1 \\ 0 & 1 & 3 & 4 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 2 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 1 & 1 & 2 & -1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 0 & 1 & 2 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow [R \quad \vec{d}]$$

Find the general solution.

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

- Next, find the special solutions of $A\vec{x} = \vec{0}$ and a particular solution of $A\vec{x} = \vec{b}$:

$$[R \quad \vec{d}] = \begin{bmatrix} 0 & 1 & 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- There are three free variables, x_1 , x_4 and x_5 , and thus three special solutions:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ ? \\ ? \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1}, \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \vec{x}_{n2}, \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -2 \\ 0 \\ 1 \end{bmatrix} = \vec{x}_{n3}, \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_p$$

Find the general solution.

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

- Finally, put it all together:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ ? \\ ? \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_{n1}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \vec{x}_{n2}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -2 \\ 0 \\ 1 \end{bmatrix} = \vec{x}_{n3}, \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ ? \\ ? \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \vec{x}_p$$

- The general solution is:

$$\vec{x} = \vec{x}_p + s_1 \vec{x}_{n1} + s_2 \vec{x}_{n2} + s_3 \vec{x}_{n3}, \forall s_i \in \mathbb{R}$$

$$\vec{x} = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 0 \\ 0 \end{bmatrix} + s_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s_2 \begin{bmatrix} 0 \\ -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + s_3 \begin{bmatrix} 0 \\ 4 \\ -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} s_1 \\ 4 - s_2 + 4s_3 \\ -1 - s_2 - 2s_3 \\ s_2 \\ s_3 \end{bmatrix}$$

Find the rank.

Find the rank of each of the following matrices.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \\ 0 & 2 & 4 \end{bmatrix}$$

Find the rank.

Reduce to echelon form, and count pivots:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \end{bmatrix} \Rightarrow \text{rank}(A) = 2$$

$$B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & 0 \\ 0 & -3 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{rank}(B) = 2$$

$$C = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \\ 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -2 & 0 \\ 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{rank}(C) = 3$$

Find the rank by inspection

Without performing elimination, identify the rank of each of the following matrices.

$$D = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$E = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$F = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Find the rank by inspection

Without performing elimination, identify the rank of each of the following matrices.

$$D = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$E = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$F = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- $\text{rank}(D) = 1$ All non-zero rows are identical.
- $\text{rank}(E) = 0$ All rows are zero.
- $\text{rank}(F) = 4$ Reduced row echelon form with 4 pivots.